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JERUSALEM COLLEGE OF ENGINEERING

(An autonomous college affiliated to Anna University, Chennai)

Academic Year 2025-26

Assessment Test-I

Semester IV

B.E. CSE, CSE (AI&ML), CSE (CS), ECE & B.Tech. AI&DS, CS&BS, IT

JMA2402-Applied Linear Algebra

Regulation 2023

TIME : 08.30 am – 10.00 am

DATE: 14.02.2026

MAX. MARKS: 50 marks

PART A (5×2=10)

Q.NO.	QUESTION	COURSE OUTCOME	BT
1	Define rank of a matrix.	CO1	K1
2	What is the condition for a system of linear equations to have (i) exactly one solution (ii) infinitely many solutions.	CO1	K1
3	Define linear dependence and independence of vectors.	CO2	K1
4	Define dimension with an example.	CO2	K1
5	If $T: V \rightarrow W$ is a linear transformation then show that $T(0) = 0$.	CO3	K2

PART B (2×16=32)

6(a)	(i) Find the values of λ for which the equations $x + 2y + z = 4$, $2x - y - z = 3\lambda$ and $4x - 7y - 5z = \lambda^2$ may have a solution. For these values of λ find the solution also.	CO1	K4
	(ii) Solve the system of equations by Gauss Jordan method. $2x_1 + x_2 + x_3 = 10$, $3x_1 + 2x_2 + 3x_3 = 18$ and $x_1 + 4x_2 + 9x_3 = 16$		
OR			
6(b)	(i) For what values of λ, μ does the system of equations $x + y + z = 6$, $x + 2y + 3z = 10$ & $x + 2y + \lambda z = \mu$ have (a) many solutions (b) no solution (c) only one solution.	CO1	K4
	(ii) Solve the system $9x_1 + 2x_2 + 4x_3 = 10$, $2x_1 - 4x_2 + 10x_3 = -15$ and $x_1 + 10x_2 + 4x_3 = 6$ by Gauss Seidel method correct to 4 decimal places.		
7(a)	Show that the set of all real valued polynomials of degree less than or equal to n denoted by $P_n(\mathbb{R})$ is a vector space with operations of addition & scalar multiplication.	CO2	K4
OR			
7(b)	(i) Prove that $W = \{(a_1, a_2, a_3) \in \mathbb{R}^3 / a_1 = 3a_2 \text{ \& } a_3 = -a_2\}$ is a subspace of the vector space $\mathbb{R}^3$ defined over the field of real numbers $\mathbb{R}$ under coordinate wise addition and scalar multiplication.	CO2	K4
	(ii) Check whether the set of vectors $\left\{ \begin{bmatrix} 8 \\ -5 \end{bmatrix}, \begin{bmatrix} 1 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$ are linearly dependent, if so express any one vector as a linear combination of other vectors.		

PART C (1×8=8)

8(a)	Check if $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as given below, are linear transformations or not. (a) $T(a_1, a_2) = (a_1, a_2^2)$ (b) $T(a_1, a_2) = (a_1 + 1, a_2)$	CO3	K4
OR			
8(b)	Define reflection about x -axis and projection on x -axis and show that they are linear transformations.	CO3	K4